

Model overview.

Scores for test set only.

Model name	Parameters we can change	Initial score	Best f1 score after GridSearchCV
Random Forest	n_estimators, criterion, max_depth	acc: 92% f1-class1: 95%	95%
SVM	kernel, gamma, C, degree	acc: 89% f1-class1: 92%	94%
XGBoost	booster, num_boost_round, max_depth	acc: 91% f1-class1: 94%	95%
KNN	n_neighbors, weights, metric	acc: 90% f1-class1: 93%	94%

Comparison between train and test set scores to address overfitting.

OS: oversample, US: undersample

f1-scores after GridSearchCV (in %)		OS_MINMAX	OS_ROBUST	US_ROBUST
Random Forest	Train	99.8	1	99.9
	Test	94.9	94.2	94.2
SVM	Train	89.3	92.9	88.8
	Test	92.3	93.8	92.8
XG BOOST	Train	99.9	99.9	96.8
	Test	94.6	93.3	93.3
KNN	Train	1	1	1
	Test	93.7	93.2	93.5

As a conclusion, we chose the SVM model with OS_ROBUST data as the best model due to its low overfitting and high f1 score.